



National
Qualifications
2014

2014 Mathematics Paper 1 (Non-calculator)

National 5

Finalised Marking Instructions

© Scottish Qualifications Authority 2014

The information in this publication may be reproduced to support SQA qualifications only on a non-commercial basis. If it is to be used for any other purposes written permission must be obtained from SQA's NQ Assessment team.

Where the publication includes materials from sources other than SQA (secondary copyright), this material should only be reproduced for the purposes of examination or assessment. If it needs to be reproduced for any other purpose it is the centre's responsibility to obtain the necessary copyright clearance. SQA's NQ Assessment team may be able to direct you to the secondary sources.

These Marking Instructions have been prepared by Examination Teams for use by SQA Appointed Markers when marking External Course Assessments. This publication must not be reproduced for commercial or trade purposes.

Detailed Marking Instructions for each question

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
1.			Ans: $\frac{25}{27}$ • ¹ start to multiply fractions • ² consistent answer in simplest form	2	• ¹ $\frac{5}{12} \times \frac{20}{9}$ or $2 \times \frac{5}{12} + \frac{2}{9} \times \frac{5}{12}$ • ² $\frac{25}{27}$
Notes: 1. Correct answer without working award 2/2. 2. $\frac{100}{108}$ (no working necessary) award 1/2. 3. 2 nd mark only available where simplifying is required. 4. For subsequent incorrect working, the final mark is not available eg $\frac{25}{27} = 1\frac{2}{27}$ award 1/2.					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
2.			Ans: $6x^2 - 13x - 5$ • ¹ any three terms correct • ² fourth term correct and collect like terms	2	• ¹ eg $6x^2 + 2x - 15x$ • ² $6x^2 - 13x - 5$
Notes: 1. Correct answer without working award 2/2					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
3.			Ans: $(x-7)^2 - 5$ • ¹ correct bracket with square • ² complete process	2	• ¹ $(x-7)^2$ • ² $(x.....)^2 - 5$
Notes: 1. For $(x-7)^2 + (-5)$, $(x-7)(x-7) - 5$ award 2/2 2. For $(x-7) - 5$, $(x^2 - 7) - 5$, $(x^2 - 7)^2 - 5$, $(x-7x)^2 - 5$ award 1/2 ✗✓					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
4.			Ans: $\begin{pmatrix} -4 \\ 10 \\ 3 \end{pmatrix}$ •¹ calculate 2u •² solution	2	•¹ $\begin{pmatrix} -4 \\ 6 \\ 10 \end{pmatrix}$ •² $\begin{pmatrix} -4 \\ 10 \\ 3 \end{pmatrix}$
Notes: 1. Correct answer without working award 2/2. 2. Brackets not required 3. For $(-4,10,3)$ award 1/2 4. For subsequent invalid working, the final mark is not available. eg $9(-4+10+3)$, $\sqrt{125}$ (magnitude) award 1/2					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
5.			Ans: 8 cm •¹ correct substitution into sine rule •² know how to solve •³ correct calculation	3	•¹ $\frac{LM}{0.4} = \frac{18}{0.9}$ •² $(LM =) \frac{0.4 \times 18}{0.9}$ •³ $(LM =) 8$
Notes: 1. For $\frac{LM}{\sin 0.4} = \frac{18}{\sin 0.9} \rightarrow \frac{18 \sin 0.4}{\sin 0.9} = 8$ award 2/3 x✓✓ 2. For $\frac{LM}{\sin 0.4} = \frac{18}{\sin 0.9} \rightarrow \frac{LM}{0.4} = \frac{18}{0.9} \rightarrow \frac{18 \times 0.4}{0.9} = 8$ award 2/3 x✓✓					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
6.	(a)		Ans: $C = 15F + 125$ Method 1: $y = mx + c$ • ¹ find gradient • ² substitute gradient and a point into $y = mx + c$ • ³ calculate c , then state equation in simplest form in terms of F and C	3	• ¹ $\frac{300}{20}$ • ² e.g. $200 = \frac{300}{20} \times 5 + c$ • ³ $C = 15F + 125$ or equivalent
			Method 2: $y - b = m(x - a)$ • ¹ find gradient • ² substitute gradient and a point into $y - b = m(x - a)$ • ³ expand brackets and rearrange equation into simplest form in terms of F and C		• ¹ $\frac{300}{20}$ • ² e.g. $y - 200 = \frac{300}{20}(x - 5)$ • ³ $C = 15F + 125$ or equivalent
Notes: 1. For correct answer without working, award 3/3 2. For $y = 15x + 125$ award 2/3 3. For $y = 15x$ award 1/3 4. Where m and/or c are incorrect the working must be followed through to give the possibility of awarding 1/3 or 2/3 5. If the equation is stated incorrectly and there is no working, 1/3 can be awarded for correct gradient or correct y -intercept 6. For an incorrect equation (ie both m and c incorrect), without working, eg $C = 125F + 15$ award 0/3					
	(b)		Ans: 725 calories • ¹ calculate value using the equation	1	• ¹ $C = 15 \times 40 + 125 = 725$
Notes: 1. For a correct answer without working award 0/1 2. Follow through mark from part (a) is only available if the calculation involves a multiplication or division and an addition or subtraction					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
7.			Ans: $a = 5$ • ¹ know to substitute $(-3, 45)$ into $y = ax^2$ • ² solve equation for a	2	• ¹ $45 = a(-3)^2$ or equivalent • ² $a = 5$
Notes: 1. For a correct answer without working 2. For $45 = a \times (-3) \rightarrow a = -15$					
					award 2/2 award 0/2

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
8.			Ans: $9\sqrt{10}$ • ¹ simplify $\sqrt{40}$ • ² simplify $\sqrt{90}$ • ³ state answer in simplest form	3	• ¹ $2\sqrt{10}$ • ² $3\sqrt{10}$ • ³ $9\sqrt{10}$
Notes: 1. For a correct answer without working 2. For subsequent incorrect working, the final mark is not available.					
					award 0/3

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
9.			Ans: 600 000 • ¹ know that $80\% = 480\,000$ • ² begin valid strategy • ³ answer	3	• ¹ $80\% = 480\,000$ • ² $10\% = 60\,000$ or equivalent • ³ 600 000
Notes: 1. For 600 000 with or without working 2. For 384 000 (80% of 480 000) or 576 000 (120% of 480 000) (i) and evidence of $80\% = 480\,000$ (ii) otherwise					
					award 3/3 award 1/3 award 0/3

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
10.			Ans: $a = 3, b = -40$ • ¹ state value of a • ² state value of b	2	• ¹ $a = 3$ • ² $b = -40$
Notes: 1. For $y = 3\sin(x - 40)$ 2. Accept $b = 320$					

award 2/2

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
11.	(a)		Ans: gradient = $-\frac{4}{3}$ • ¹ start to rearrange • ² state gradient	2	• ¹ $3y = -4x + 12$ • ² $-\frac{4}{3}$

Notes:

1. Correct answer without working award 2/2

2. Some common answers (no working necessary)

(a) $-1.\dot{3}, -1.33$ award 2/2

(b) -1.3 award 1/2

(c) $-\frac{4}{3}x$ award 1/2

(d) $\frac{4}{3}$ award 1/2

(e) $\frac{4}{3}x$ award 0/2

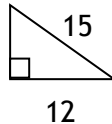
	(b)		Ans: (3,0) • ¹ know how to find x-coordinate • ² state coordinates (must use brackets)	2	• ¹ $4x + 3(0) = 12$ or equivalent • ² (3,0)
--	-----	--	---	---	---

Notes:

1. For (3,0) without working award 2/2

2. For $x=3$ with or without working award 1/2

3. For (0,4) with or without working award 1/2

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
12.			Ans: 18 centimetres • ¹ marshal facts and recognise right angle • ² know how to use Pythagoras • ³ correct calculation of PA^2 • ⁴ find length of PQ	4	• ¹  • ² $x^2 = 15^2 - 12^2$ • ³ 81 • ⁴ 18

Notes:

1. For 18 without valid working award 0/4

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
13.	(a)		Ans: 6 seconds • ¹ construct an equation • ² rearrange and equate to zero • ³ correct factorisation • ⁴ solve equation and select correct value	4	• ¹ $16t - t^2 = 60$ • ² eg $t^2 - 16t + 60 = 0$ • ³ $(t - 6)(t - 10)$ • ⁴ $(t =) 6$
Notes: 1. Equating to zero must appear prior to solving equation e.g. $t^2 - 16t + 60 \rightarrow (t - 6)(t - 10) \rightarrow (t =) 6$ award 3/4 2. For the case in Note 1, if 6 is not stated explicitly award 1/4 3. For an answer of 6 without working award 0/4 4. Where a candidate substitutes into the formula (a) $h(6)=60$ and $h(10)=60 \rightarrow 6$ award 4/4 (b) $h(6)=60 \rightarrow 6$ award 2/4 (c) $h(6)=60$ award 1/4 (d) $h(10)=60 \rightarrow 10$ award 1/4					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
13.	(b)		Ans: No, because its maximum height is 64 metres. <u>Method 1</u> •¹ know that turning point is at 8 seconds •² calculate maximum height •³ communicate conclusion <u>Method 2</u> •¹ calculate $h(8)$ •² demonstrate that $h(8)$ is maximum height •³ communicate conclusion <u>Method 3</u> •¹ evidence of using $b^2 - 4ac$ in $16t - t^2 = 70$ •² demonstrate that $b^2 - 4ac < 0$ •³ communicate conclusion	3	•¹ $t = 8$ (stated or implied) •² $16(8) - 8^2 = 64$ •³ no, it only reaches 64 metres. •¹ 64 •² e.g. $h(7)=63$ and $h(9)=63$ •³ no, it only reaches 64 metres. •¹ evidence •² e.g. $256 - 280 < 0$ •³ no, as equation has no real roots.
Notes: 1. For final mark (Methods 1 and 2), answer must include valid comparison or an implied comparison eg 'only' or 'less than'. eg No, it reaches 64 metres award 2/3 No, it reaches 64 metres, $64 < 70$ award 3/3 2. Where a trial and improvement method is used (i) accept trials appearing in parts (a) and (b) (ii) accept scored out working as evidence of rejected trials					

[END OF MARKING INSTRUCTIONS]



National
Qualifications
2014

2014 Mathematics Paper 2

National 5

Finalised Marking Instructions

© Scottish Qualifications Authority 2014

The information in this publication may be reproduced to support SQA qualifications only on a non-commercial basis. If it is to be used for any other purposes written permission must be obtained from SQA's NQ Assessment team.

Where the publication includes materials from sources other than SQA (secondary copyright), this material should only be reproduced for the purposes of examination or assessment. If it needs to be reproduced for any other purpose it is the centre's responsibility to obtain the necessary copyright clearance. SQA's NQ Assessment team may be able to direct you to the secondary sources.

These Marking Instructions have been prepared by Examination Teams for use by SQA Appointed Markers when marking External Course Assessments. This publication must not be reproduced for commercial or trade purposes.

Detailed Marking Instructions for each question

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
1.			Ans: 590 • ¹ know how to decrease by 15% • ² know how to calculate roll • ³ carry out calculations correctly within a valid strategy and round to the nearest ten	3	• ¹ $\times 0.85$ • ² 964×0.85^3 • ³ 590
Notes: 1. For an answer of 590 without working award 3/3 ✓✓✓ 2. For an answer of 592 or 592.0165 without working award 2/3 ✓✓x 3. Where an incorrect percentage has been used, the working must be followed through to give the possibility of awarding 2/3 x✓✓ 4. For an answer of 2460 ($964 \times 0.85 \times 3$) with working, award 1/3 ✓xx 5. For an answer of 530 ($964 - 964 \times 0.15 \times 3$) with working, award 1/3 ✓xx 6. For an answer of 430 ($964 \times 0.15 \times 3$) award 0/3 xxx					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
2.			Ans: B (8, 4, 10), C (4, 0, 10) • ¹ state coordinates of B • ² state coordinates of C	2	• ¹ (8, 4, 10) • ² (4, 0, 10)
Notes: 1. For eg B(8, 4, 9) leading to C(4, 0, 9) award 1/2 x✓ 2. The maximum mark available is 1/2 where (a) brackets are omitted (b) answers are given in component form					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
3.	(a)		Ans: $5a + 3c = 158.25$ • ¹ construct equation	1	• ¹ $5a + 3c = 158.25$
Notes: 1. Accept variables other than a and c .					
	(b)		Ans: $3a + 2c = 98$ • ¹ construct equation	1	• ¹ $3a + 2c = 98$
Notes:					
	(c)		Ans: Adult ticket costs £22.50 Child ticket costs £15.25 • ¹ evidence of scaling • ² follow a valid strategy through to produce values for a and c • ³ calculate correct values for a and c • ⁴ communicate answers in money	4	• ¹ eg $10a + 6c = 316.50$ $9a + 6c = 294$ • ² values for a and c • ³ $a = 22.5$ and $c = 15.25$ • ⁴ Adult £22.50 Child £15.25
Notes: 1. The fourth mark may only be awarded when all of the following are given in the final answer: the words “adult” and “child”, the £ signs and both amounts written with two decimal figures.					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •						
4.	(a)	(i)	Ans: $\bar{x} = 56.5$ • ¹ calculate mean	1	• ¹ $\bar{x} = 56.5$						
Notes: 1. Do not accept 56.5 rounded to 57.											
		(ii)	Ans: $s = 2.4$ • ¹ calculate $(x - \bar{x})^2$ • ² substitute into formula • ³ calculate standard deviation	3	• ¹ 0.25, 0.25, 2.25, 2.25, 12.25, 12.25 • ² $\sqrt{\frac{29.5}{5}}$ • ³ 2.4(2....)						
Notes: 1. For use of alternative formula, award marks as follows: <table style="width: 100%; border: none;"> <tr> <td style="width: 50%;">•¹ calculate $\sum x$ and $\sum x^2$</td> <td style="width: 50%;">•¹ 339, 19183</td> </tr> <tr> <td>•² substitute into formula</td> <td>•² $\sqrt{\frac{19183 - \frac{339^2}{6}}{5}}$</td> </tr> <tr> <td>•³ calculate standard deviation</td> <td>•³ 2.4(2....)</td> </tr> </table>						• ¹ calculate $\sum x$ and $\sum x^2$	• ¹ 339, 19183	• ² substitute into formula	• ² $\sqrt{\frac{19183 - \frac{339^2}{6}}{5}}$	• ³ calculate standard deviation	• ³ 2.4(2....)
• ¹ calculate $\sum x$ and $\sum x^2$	• ¹ 339, 19183										
• ² substitute into formula	• ² $\sqrt{\frac{19183 - \frac{339^2}{6}}{5}}$										
• ³ calculate standard deviation	• ³ 2.4(2....)										
2. For correct answer without working award 0/3											
	(b)		Ans: No, standard deviation is greater OR No, times are more spread out • ¹ no, with valid explanation	1	• ¹ e.g. No, standard deviation is greater						
Notes: 1. Answer must be consistent with answer to part (a)(ii). 2. Accept “No, as $3.2 > 2.4$ ” 3. Only award the mark if it is clear that the reason is based on standard deviation only . 4. Do not accept “No, times are less consistent” without further explanation.											

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
5.			Ans: 3072 cm³ • ¹ state linear scale factor • ² state volume scale factor • ³ calculate volume (calculation must involve a power of the scale factor) and state correct units	3	• ¹ $\frac{24}{15}$ or 1.6 • ² $\left(\frac{24}{15}\right)^3$ or 1.6^3 (= 4.096) • ³ 3072cm ³

Notes:

1. Some common answers

(a) 3072 award 2/3 ✓✓x

(b) 1920cm³ ($\left(\frac{24}{15}\right)^2 \times 750$) award 2/3 ✓x✓

(c) 1200cm³ ($\left(\frac{24}{15}\right) \times 750$) award 1/3 ✓xx

(d) 675000000cm³ ($\left(\frac{24}{15}\right) \times 750^3$) award 1/3 ✓xx

(e) 183cm³ ($\left(\frac{15}{24}\right)^3 \times 750$) award 2/3 x✓✓

(f) 933cm³ ($\left(\frac{15}{24}\right)^3 \times 750 + 750$) award 2/3 x✓✓

2. The third mark is not available where premature rounding leads to an incorrect answer.
 eg $4.1 \times 750 = 3075\text{cm}^3$ award 2/3 ✓✓x

3. Alternative Method

- | | |
|--|---|
| • ¹ know how to find radius of smaller cylinder | • ¹ $\sqrt{\frac{750}{15\pi}}$ |
| • ² know how to find radius of larger cylinder | • ² $\left(\frac{24}{15}\right) \times \sqrt{\frac{750}{15\pi}}$ |
| • ³ calculate volume and state correct units | • ³ 3072cm ³ |

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
6.			Ans: no, with valid reason. •¹ valid strategy •² evaluation •³ comparison •⁴ valid conclusion	4	•¹ use Converse of Pythagoras' Theorem eg 110^2 and $85^2 + 75^2$ •² 12 100 and 12 850 •³ e.g. $110^2 \neq 75^2 + 85^2$ •⁴ No, since not right angled

Notes:

1. For alternative methods, award marks as follows:

- | | |
|--|--|
| <p>(i) •¹ valid strategy</p> <p>•² evaluation</p> <p>•³ comparison</p> <p>•⁴ valid conclusion</p> | <p>•¹ use Pythagoras' Theorem
eg $85^2 + 75^2$</p> <p>•² $h = 113.36$</p> <p>•³ e.g. $113.36 > 110$</p> <p>•⁴ No, since not right angled</p> |
| <p>(ii) •¹ valid strategy</p> <p>•² evaluation</p> <p>•³ comparison</p> <p>•⁴ valid conclusion</p> | <p>•¹ substitute correctly into cosine rule
eg $\frac{85^2 + 75^2 - 110^2}{2 \times 85 \times 75}$</p> <p>•² 86.6°</p> <p>•³ $86.6^\circ < 90^\circ$</p> <p>•⁴ No, since not right angled</p> |

2. There must be an explicit comparison for the award of the third mark.

eg $\sqrt{85^2 + 75^2} = 113.36$.

No, since not right angled.

award 3/4 ✓✓x✓

3. Conclusion **must** involve reference to “not a right angle”.

eg $110^2 = 85^2 + 75^2 \rightarrow 12100 \neq 12850$.

No, Hightown is not due north of Lowtown.

award 3/4 ✓✓✓x

4. The final mark is not available where the candidate's only conclusion is an invalid statement involving the word bearing.

eg “No, Hightown is on a bearing of 87° from Lowtown, not 90° ”

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
7.			Ans: 150 cm³ •¹ substitute correctly into formula for volume of cone •² substitute correctly into formula for volume of sphere or hemisphere •³ know to subtract volume of hemisphere from volume of cone •⁴ carry out all calculations correctly (must involve difference or sum of two volume calculations) •⁵ round final answer to 2 significant figures	5	•¹ $\frac{1}{3} \times \pi \times 4^2 \times 15$ (= 251.32....) •² $\frac{4}{3} \times \pi \times 3 \cdot 7^3$ (= 212.17....) or $\frac{1}{2} \times \frac{4}{3} \times \pi \times 3 \cdot 7^3$ (= 106.08....) •³ evidence •⁴ 145.24.... •⁵ 150 (cm³)

Notes:

1. Accept variations in π .

2. Some common answers (working must be shown):

- | | | | |
|--------|---|-----------|-------|
| (i) | $39 \left(\frac{1}{3} \times \pi \times 4^2 \times 15 - \frac{4}{3} \times \pi \times 3 \cdot 7^3 \right)$ | award 4/5 | ✓✓x✓✓ |
| (ii) | $120 \left(\frac{1}{3} \times \pi \times 4^2 \times 15 - \frac{1}{2} \times \frac{4}{3} \times \pi \times 4^3 \right)$ | award 4/5 | ✓x✓✓✓ |
| (iii) | $110 \left(\frac{1}{3} \times \pi \times 3 \cdot 7^2 \times 15 - \frac{1}{2} \times \frac{4}{3} \times \pi \times 3 \cdot 7^3 \right)$ | award 4/5 | x✓✓✓✓ |
| (iv) | $160 \left(\frac{1}{3} \times \pi \times 8^2 \times 15 - \frac{1}{2} \times \frac{4}{3} \times \pi \times 7 \cdot 4^3 \right)$ | award 4/5 | x✓✓✓✓ |
| (v) | $360 \left(\frac{1}{3} \times \pi \times 4^2 \times 15 + \frac{1}{2} \times \frac{4}{3} \times \pi \times 3 \cdot 7^3 \right)$ | award 4/5 | ✓✓x✓✓ |
| (vi) | $460 \left(\frac{1}{3} \times \pi \times 4^2 \times 15 + \frac{4}{3} \times \pi \times 3 \cdot 7^3 \right)$ | award 4/5 | ✓✓x✓✓ |
| (vii) | $80 \left(\frac{1}{3} \times \pi \times 3 \cdot 7^2 \times 15 - \frac{1}{2} \times \frac{4}{3} \times \pi \times 4^3 \right)$ | award 3/5 | xx✓✓✓ |
| (viii) | $250 \left(\frac{1}{3} \times \pi \times 4^2 \times 15 \right)$ | award 2/5 | ✓xxx✓ |

3. The final mark is only available where answers to all intermediate steps involve at least three significant figures.

eg $251.32 - 106.08 = 250 - 110 = 140$ award 4/5 ✓✓✓✓x

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
8.			Ans: $5n^4$ • ¹ simplify powers in numerator • ² cancel constants • ³ eliminate n from denominator	3	• ¹ $10n^6$ • ² $\frac{5n^6}{n^2}$ • ³ $5n^4$

Notes:

1. For $5n^4$ without working award 3/3
2. For a final answer of $\frac{5n^4}{1}$ award 2/3 ✓×✓
3. For an answer of $5n^3$
 - (a) (i) $\frac{10n^5}{2n^2} = 5n^3$ award 2/3 ×✓✓
 - (ii) $\frac{10n^6}{2n^2} = 5n^3$ award 2/3 ✓✓×
 - (b) (i) $\frac{n^4 \times 10}{2n} = \frac{n^4 \times 10}{2} = 5n^3$ award 1/3 ✓×✓
 - (ii) $5n^3$ without working award 1/3

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
9.			Ans: $\frac{4x-15}{x(x+5)}$ • ¹ correct common denominator (or correct numerator) • ² consistent numerator (or denominator) • ³ simplify	3	• ¹ $x(x+5)$ or $7x-3(x+5)$ • ² $\frac{7x-3(x+5)}{x(x+5)}$ • ³ $\frac{4x-15}{x(x+5)}$

Notes:

1. Correct answer without working award 3/3
2. For $\frac{7x}{x(x+5)} - \frac{3(x+5)}{x(x+5)}$ award 2/3 ✓✓×
3. For subsequent incorrect working, the final mark is not available.

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
10.	(a)		Ans: 84.8° • ¹ substitute correctly into cosine rule • ² calculate cos B correctly • ³ calculate angle ABC correctly	3	• ¹ $\cos B = \frac{8^2 + 11^2 - 13^2}{2 \times 8 \times 11}$ • ² $\cos B = 0.09\ldots\ldots$ • ³ 85 or 84.8.....

Notes:

- For 1.48 (uses RAD) or 94.2 (uses GRAD), with working award 3/3
- The 2nd mark can be awarded for $\cos^{-1}\left(\frac{16}{176}\right)$

	(b)		Ans: 155.2° • ¹ know how to calculate the angle • ² correctly calculate the angle within a valid strategy	2	• ¹ $360 - 120 - [\text{answer to (a)}]$ or equivalent • ² 155.2
--	-----	--	--	---	---

Notes:

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
11.			Ans: $a = \frac{2(s-ut)}{t^2}$ • ¹ subtract ut • ² multiply by 2 • ³ divide by t^2	3	• ¹ $s - ut = \frac{1}{2}at^2$ • ² $2(s - ut) = at^2$ • ³ $a = \frac{2(s-ut)}{t^2}$

Notes:

- Correct answer without working award 3/3
- For subsequent incorrect working, the final mark is not available.
- For $a = \frac{s-ut}{\frac{1}{2}t^2}$ award 2/3

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
12.			Ans: $x^\circ = 63^\circ, 297^\circ$ •¹ rearrange equation •² find one value of x •³ find another value of x	3	•¹ $\cos x = \frac{5}{11}$ •² $x = 63$ •³ $x = 297$
Notes: 1. The 2 nd angle must be consistent with the first angle. 2. For $x = 1.1, 358.9$ (uses RAD), award 3/3 (with working), award 2/3 (without working) 3. For $x = 70, 290$ (uses GRAD), award 3/3 (with working), award 2/3 (without working)					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
13.			Ans: 151.3 m^2 •¹ know how to find area of segment •² know to express sector as a fraction of a circle •³ know how to find area of sector •⁴ know how to calculate area of triangle •⁵ carry out all calculations correctly within a valid strategy	5	•¹ evidence of e.g. major sector + triangle or circle – minor sector + triangle •² $\frac{310}{360}$ or $\frac{50}{360}$ •³ $\frac{310}{360} \times \pi \times 7^2$ (= 132.56) or $\frac{50}{360} \times \pi \times 7^2$ (= 21.38) •⁴ $\frac{1}{2} \times 7 \times 7 \times \sin 50$ (=18.77) •⁵ 151.3 m^2

Notes for question 13 are on next page.

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
Notes:			
1.	Accept variations in π ; disregard premature or incorrect rounding of $\frac{310}{360}$ or $\frac{50}{360}$.		
2.	Use of RAD or GRAD (working must be shown)		
	(a) For 149.9 [uses GRAD]		award 5/5
	(b) Where the use of RAD leads to an answer of $126.1(-6.43+132.56)$ or $139.0(6.43+132.56)$		award 4/5
3.	Some common answers (working must be shown):		
	$56.6 \left(\frac{310}{360} \times \pi \times 14 + \frac{1}{2} \times 7 \times 7 \times \sin 50^\circ \right)$	award 4/5	✓✓x✓✓
	$40.1 \left(\frac{50}{360} \times \pi \times 7^2 + \frac{1}{2} \times 7 \times 7 \times \sin 50^\circ \right)$	award 4/5	x✓✓✓✓
	$2.6 \left(\frac{50}{360} \times \pi \times 7^2 - \frac{1}{2} \times 7 \times 7 \times \sin 50^\circ \right)$	award 4/5	x✓✓✓✓
	$24.9 \left(\frac{50}{360} \times \pi \times 14 + \frac{1}{2} \times 7 \times 7 \times \sin 50^\circ \right)$	award 3/5	x✓x✓✓
	$132.6 \left(\frac{310}{360} \times \pi \times 7^2 \right)$	award 2/5	x✓✓xx
	$21.4 \left(\frac{50}{360} \times \pi \times 7^2 \right)$	award 2/5	x✓✓xx
	$18.8 \left(\frac{1}{2} \times 7 \times 7 \times \sin 50^\circ \right)$	award 1/5	xxx✓x
	$153.9 \left(\pi \times 7^2 \right)$	award 0/5	
4.	The fifth mark is only available when the area of triangle MON is calculated using trigonometry.		

[END OF MARKING INSTRUCTIONS]

General Marking Principles for National 5 Mathematics

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this Paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

- (a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.
- (b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.
- (c) Credit must be assigned in accordance with the specific assessment guidelines.
- (d) Candidates may use any mathematically correct method to answer questions except in cases where a particular method is specified or excluded.
- (e) Working subsequent to an error must be followed through, with possible credit for the subsequent working, provided that the level of difficulty involved is approximately similar. Where, subsequent to an error, the working is easier, candidates lose the opportunity to gain credit.
- (f) Where transcription errors occur, candidates would normally lose the opportunity to gain a processing mark.
- (g) Scored out working which **has not been replaced** should be marked where still legible. However, if the scored out working **has been replaced**, only the work which has not been scored out should be marked.
- (h) Where a candidate has made multiple attempts, mark all attempts and award the lowest mark.
- (i) Unless specifically mentioned in the specific assessment guidelines, do not penalise:
 - Working subsequent to a correct answer
 - Correct working in the wrong part of a question
 - Legitimate variations in solutions
 - Bad form
 - Repeated error within a question